Title:

"Early Detection of Stroke Using Machine Learning and Computer Vision"

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Data collecting - building ML model - deployment

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Introduction

problem definition

Stroke is one of the leading causes of death and long-term disability in the world. It occurs due to interruption of blood supply to the brain, leading to extensive damage in brain cells. The challenge with stroke is that it is time-dependent: the sooner it is diagnosed and treated, the higher the chances of healing and survival. However, the traditional diagnosis procedures are predominantly dependent on manual image reading (e.g., CT or MRI scans) by doctors, a labor-intensive and error-prone procedure. Besides, in the majority of areas in the world, especially in developing countries, doctors and radiologists of required level are not available there, thus subjecting the patient to delayed and wrong diagnosis. This creates an emergent requirement for an automated, accurate, and effective system that can assist doctors in recognizing stroke cases at early stages.

Importance

Developing an AI system for stroke diagnosis can

potentially create significantly improved healthcare outcomes. With the integration of machine learning and computer vision techniques, medical images can be rapidly and accurately analyzed to recognize stroke indications. This kind of system could help doctors make more precise and rapid decisions, reduce the workload of medical personnel, and rule out the chances of misdiagnosis. Moreover, early detection with the help of AI can reduce death rates, improve recovery among patients, and reduce the cost of healthcare for stroke complications. This project aims to create and implement an intelligent diagnostic model that complements traditional medical practice with the wisdom of artificial intelligence, thus contributing to enhanced healthcare efficiency and availability.

Methodology for the solution

1. Data Collection & Preprocessing

Gather CT/MRI brain scan datasets and enhance image quality using preprocessing techniques.

2. Feature Extraction

Apply computer vision or deep learning methods (e.g., CNNs) to extract relevant features from brain images.

3. Model Training

Train machine learning/deep learning models to classify cases into stroke and non-stroke.

Use data augmentation and split the dataset into training, validation, and testing sets.

4. Evaluation

Assess model performance using accuracy, precision, recall, and F1-score.

5. System Deployment

Build a simple interface for doctors to upload scans and get automated diagnosis with visual explanations.

Area of Study

The focus of this project is Artificial Intelligence in Healthcare, specifically using Machine Learning and Computer Vision for stroke diagnosis. The study is focused on the application of medical images (CT and MRI) to study cases of stroke detection and classification.

This area of study crosses computer science and medicine in an aim to develop intelligent systems that help doctors with early and accurate diagnosis, which improves healthcare quality as well and reduce stroke mortality.

Expected Outcomes

1. Stroke Diagnosis System

A functional AI-based model capable of classifying brain scans into stroke and non-stroke cases.

2. High Diagnostic Accuracy

The system is expected to achieve high accuracy,

precision, and recall, reducing the risk of misdiagnosis.

3. Time Efficiency

Faster analysis of CT/MRI scans compared to manual diagnosis by doctors.

4. User-Friendly Interface

A simple platform that allows medical staff to upload scans and receive automated diagnostic results.

5. Support for Medical Experts

The system will act as a decision-support tool, assisting doctors rather than replacing them.

Required tools

Software

Programming Language: Python.

Libraries & Frameworks:

TensorFlow / Keras or PyTorch (for deep learning models).

OpenCV (for image preprocessing and computer vision tasks).

NumPy, Pandas, Matplotlib (for data handling and visualization).

Scikit-learn (for classical ML algorithms and evaluation).

Development Environment: Jupyter Notebook / Google Colab / VS Code.

Datasets

Kaggle Stroke Prediction Dataset (for additional clinical data).

Time plan

Week 1,2 : data preprocessing
Week 3,4 : building and training ML model
Week 5,6 : testing and improve the accuracy
Week 7,8 : training computer vision model
Week 9 : testing and improve the accurate
Week 10,11: build simple interface and deploy the project

References

Kaggle